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MAE 3240 HW2

2/10

1)

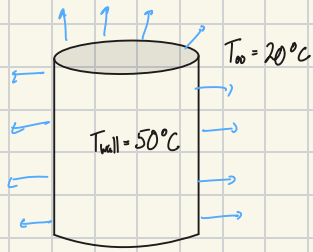
Known

- $R = 0.5\,m$, $H = 1\,m$, $T_{wall} = 50°C$
- $T_\infty = 20°C$, $h = 15\ \frac{W}{m^2 \cdot K}$
- $L = .02\,m$, $k = 0.1\ \frac{W}{m \cdot K}$

Find

- required electrical power no foam & foam
- impacts of adding foam on energy needed to insulate

Schematic



Assumptions

- Electrical energy in = heat out
- Bottom is well insulated
- Radiation is negligible

Analysis

a) $A_{tot} = A_{side} + A_{top} = 2\pi(0.5m)(1m) + \pi(0.5m)^2$

$= 3.93\,m^2$

$Q = h \cdot A \cdot \Delta T = (15\ \frac{W}{m^2 \cdot K}) \cdot (3.93\,m^2)(30K) = 1.77\,kW$

b) $R_{cond} = \frac{L}{k} = 0.2$ $\quad R_{conv} = \frac{1}{h} = 0.067$

$R_{tot} = 0.267\ \frac{m^2 \cdot K}{W}$

$q_{foam} = \frac{\Delta T}{R_{tot}} = 112.5\ W/m^2 K$

$q_{tot} = q_{foam} \cdot A_{tot} = 442\,W$

c) $\frac{q_{tot}}{q_{no\,foam}} = 0.26 \longrightarrow$ same ΔT produces smaller heat loss $\rightarrow$ less electrical input needed

2)

Known

- $L = 0.1\,m$, $k = 1\ \frac{W}{m \cdot K}$
- $T_\infty = 20°C$, $h = 100\ \frac{W}{m^2 \cdot K}$
- $q = 1000\ \frac{W}{m^2}$
- $T_{max} = 200°C$

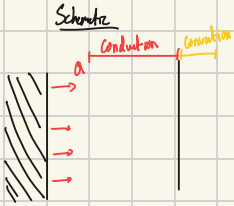
Find

- $T(0)$, $T(L)$
- q_{max} @ $T(0) < T_{max}$
- How to increase q_{max} @ $T(0) < T_{max}$

Assumptions

- no radiation, left side insulated
- Steady-state, constant k

Schematic



Analysis

a) $q_{convection} = h\Delta T = h[T(L) - T_\infty]$

$\hookrightarrow T_L = T_\infty + \frac{q}{h} = 30°C$

$\frac{k \cdot \Delta T}{L} = q \rightarrow \frac{1(T_0 - 30°C)}{0.1m} = 1000\ \frac{W}{m^2 \cdot K} \longrightarrow T_0 = 130°C$

b) $R_{tot} = \frac{L}{k} + \frac{1}{h} = 0.1 + 0.01 = 0.11$

$\Delta T_{max} = 180K$

$q_{tot} = \frac{\Delta T_{max}}{R_{tot}} = \frac{180K}{0.11} = 1.64\ \frac{kW}{m^2}$

c) higher h, higher k, decrease L

higher heat flow — more conduction transfer — thinner layer for transfer

3)

Known

- given isotherm

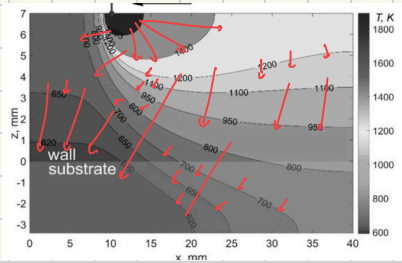
Find

- heat flux vectors

Assumptions

- 2-D heat transfer in x & z directions

Schematic + a.)



Analysis

$\vec{q} = -k \nabla T$

↳ moves from hot to cold

↳ perpendicular due to gradient

b) there is tighter contour spacing on the left side - so heat transfer gradient is greater